

Khulna University of Engineering & Technology
Department of Building Engineering and Construction Management
B. Sc. Engineering 2nd Year 1st Term Regular Examination, 2020
Math 2123
(Mathematics III)

Full Marks: 120

Time: 1.5 hrs

- N.B.
- Answer any two questions from each section in separate script.
 - Figures in the right margin indicate full marks
 - Assume reasonable values for any missing data.

Section - A

- Find the value of λ so that the vectors $\lambda\hat{i} - \hat{j} + \hat{k}$, $\hat{i} + 2\hat{j} - 3\hat{k}$ and $3\hat{i} - 4\hat{j} + 5\hat{k}$ are not coplanar. (10)
 - Find $\vec{a} \times (\vec{b} \times \vec{c})$ and $(\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$ where $\vec{a} = 2\hat{i} + 3\hat{j} + 4\hat{k}$, $\vec{b} = \hat{i} + \hat{j} - \hat{k}$, $\vec{c} = \hat{i} - \hat{j} + \hat{k}$. Also find the angle between $\vec{a} \times (\vec{b} \times \vec{c})$ and $\vec{b} \times \vec{c}$. (20)
- The derivative of $f(x, y)$ at $P(1, 2)$ in the direction of $\hat{i} + \hat{j}$ is $2\sqrt{2}$ and in the direction of $-\hat{i} - 2\hat{j}$ is -3 . What is the derivative of $f(x, y)$ in the direction of $-\hat{i} - 2\hat{j}$? Give reasons for your answer. (18)
 - Find the work done in a moving particle in the force field, $\vec{F} = 3x^2\hat{i} + (2xz - y)\hat{j} + z\hat{k}$ along the curve defined by $x^2 = 4y$, $3x^3 = 8z$ from $x = 0$ to $x = 2$. (12)
- Let $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ then find all eigen values and corresponding eigen vectors. (10)
 - The augmented matrix of a linear system has the form $\begin{bmatrix} a & 1 & : & 1 \\ 2 & (a-1) & : & 1 \end{bmatrix}$
 - Determine the values of a for which the linear system is consistent. (14)
 - When it is consistent, does the system have a unique solution or infinitely many solutions?
- Find the directional derivatives of $\phi = x^2yz + 4xz^2$ at $(1, -2, -1)$ in the direction $2\hat{i} - \hat{j} - 2\hat{k}$. (06)

Section - B

- Define Laplace transform. Find $L\{f(t)\}$ where, $f(t) = \begin{cases} \sin t & 0 \leq t < \pi/4 \\ \sin t + \cos(t - \pi/4) & t \geq \pi/4 \end{cases}$ (10)
- Find the Inverse Laplace transform, $L^{-1}\left\{\frac{1}{(s+2)^2(s-2)}\right\}$ (10)
- Find the value of Integral, $\int_0^{\infty} t^3 e^{-t} \sin t dt$

$$\begin{aligned} \mathcal{L}\{f(t)g(t)\} &= F(s)G(s) \\ \mathcal{L}\{f(t)g(t)\} &= \int_0^t f(u)g(t-u)du \\ &= F(s)G(s) \end{aligned}$$

5. (a) Using Laplace transform solve the following integral equation, (15)

$$F(t) = 1 + 2 \int_0^t F(t-x)e^{-2x}dx$$

- (b) Using Laplace transform solve the differential equation: (15)

$$y^{(iv)}(t) - 16y(t) = 30\sin t$$

subject to $y(0) = 0, y'(0) = 2, y''(\pi) = 0, y'''(\pi) = -18$.

6. (a) Define adjoint of a matrix. Find the inverse of (15)

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 5 & 7 & 8 \\ 4 & 10 & 12 \end{bmatrix}$$

- (b) Find the rank of the following real matrix: (15)

$$\begin{bmatrix} a & 1 & 2 \\ 1 & 1 & 1 \\ -1 & 1 & 1-a \end{bmatrix}$$

Where 'a' is a real number.

$$\sim \begin{bmatrix} 1 & 1 & 2 \\ a & 1 & 2 \\ -1 & 1 & 1-a \end{bmatrix} \quad \begin{matrix} (1-a)(2-a) \\ 2-3a+a^2-4+12a \end{matrix}$$

$$\begin{matrix} R_2 \rightarrow R_2 - aR_1 \\ R_3 \rightarrow R_3 + R_1 \end{matrix} \quad \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1-a & 2-a \\ 0 & 2 & 2-a \end{bmatrix}$$

$$R_3 \rightarrow R_3 - (1-a)R_2 + 2R_2$$

$$= \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1-a & 2-a \\ 0 & 0 & a^2-a-2 \end{bmatrix}$$

$$\text{Rank} = 3 \quad |a \neq 2, -1$$

$$\text{if } a \neq 1, 2$$

$$\therefore \text{Rank} = 3$$

$$K=2, \text{Rank } 2$$

$$\text{or } K=-1 \text{ or } 2$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1-a & 2-a \\ 0 & 0 & (a-2)(a+1) \end{bmatrix}$$

$$\text{Rank} = 3$$

$$\text{if } a \neq 2, -1$$

$$\text{Rank} = 2$$

$$\text{if } a = 2, -1$$

$$\frac{(k-2)(k+1)}{k^2-k-2}$$

$$\int_0^t f(t-x)g(x)dx = \int_0^t f(x)g(t-x)dx$$

$$F(t) = 1 + 2 \int_0^t F(t-x)e^{-2x}dx$$

$$F(t) = 1 + 2 \left[F(t) \frac{1-e^{-2t}}{2} \right]$$

$$f(s) = \frac{1}{s} + 2 \left[f(s) \frac{1}{s+2} \right]$$

$$\text{Laplace transform} \rightarrow f(s) = \frac{1}{s} + \frac{2f(s)}{s+2}$$

$$(s+2)f(s) = \frac{s+2}{s} + 2f(s)$$

$$f(s) = \frac{1}{s} + \frac{2}{s+2}$$

$$\text{inverse Laplace} \rightarrow F(t) = 1 + 2e^{-2t}$$